WORKSHEET 8.2

Position, Velocity and Acceleration by Integration

Unit 8: Applications of Integration

Key ideas

- **Displacement** on $[a, b]$ is $\displaystyle\int_a^b v(t)\,dt$. **Position**: $x(b) = x(a) + \displaystyle\int_a^b v(t)\,dt$.
- **Total distance** is $\displaystyle\int_a^b |v(t)|\,dt$. Without a calculator, split the interval where v changes sign.
- **Velocity**: $v(b) = v(a) + \displaystyle\int_a^b a(t)\,dt$.
- Speed is $|v|$. It increases when v and a have the same sign.

Worked examples

Example 1 · Position

A particle has $v(t) = t^2 - 6t + 8$ and $x(0) = 1$. Find $x(6)$.

Solution. $\displaystyle\int_0^6 (t^2 - 6t + 8)\,dt = \left[\tfrac{t^3}{3} - 3t^2 + 8t\right]_0^6 = 12.$

$x(6) = 13$.

Example 2 · Total distance

Find the total distance travelled by that particle for $0 \le t \le 6$.

Solution. $v = 0$ at $t = 2$ and $t = 4$. The displacements on the three intervals are $\dfrac{20}{3}$, $-\dfrac{4}{3}$ and $\dfrac{20}{3}$.

$\dfrac{20}{3} + \dfrac{4}{3} + \dfrac{20}{3} = \dfrac{44}{3}$.

Example 3 · From acceleration

A particle has $a(t) = 6t - 4$ and $v(0) = 3$. Find $v(2)$.

Solution. $v(2) = 3 + \displaystyle\int_0^2 (6t - 4)\,dt = 3 + (12 - 8).$

$v(2) = 7$.

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Practice

CORE

1. A particle has $v(t) = 4t - 6$. Find its displacement for $0 \leq t \leq 5$.

2. For the particle in Question 1, find the total distance travelled for $0 \leq t \leq 5$.

3. A particle has $a(t) = 12t^2$ and $v(0) = -5$. Find $v(2)$.

4. A particle has $x(1) = 4$ and $\displaystyle\int_1^6 v(t)\,dt = -9$. Find $x(6)$.

EXAM LEVEL

5. A particle has $v(t) = 6\sin(2t)$ for $0 \leq t \leq \pi$. Find its displacement and the total distance travelled.

6. A particle has $v(t) = e^t - 3$ and $x(0) = 5$. Find its position at the moment it is at rest.

7. A ball is thrown upward with velocity 20 metres per second, and its acceleration is -10 metres per second per second. How far above its starting point does it rise?

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8. *Calculator.* A particle has $v(t) = e^{t/2}\cos t$ for $0 \le t \le 4$. Find the total distance travelled, and the displacement.

Question 9 · Free-response practice

A particle moves along the x-axis with velocity $v(t) = t^3 - 6t^2 + 8t$ for $0 \le t \le 5$. At time $t = 0$ the particle is at $x = 3$.

(a) Find the interval of time during which the particle is moving to the left. Justify your answer.

(b) Find the position of the particle at $t = 4$, and the total distance it travels for $0 \le t \le 4$.

(c) Is the speed of the particle increasing or decreasing at $t = 3$? Give a reason.

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Worked solutions

1. $\Big[2t^2 - 6t\Big]_0^5 = 20$.

2. $v = 0$ at $t = 1.5$. Displacements -4.5 and 24.5: distance 29.

3. $-5 + \Big[4t^3\Big]_0^2 = 27$.

4. $4 + (-9) = -5$.

5. Displacement $\Big[-3\cos(2t)\Big]_0^{\pi} = 0$. Distance $2\displaystyle\int_0^{\pi/2} 6\sin(2t)\,dt = 12$.

6. At rest when $t = \ln 3$. $x = 5 + \Big[e^t - 3t\Big]_0^{\ln 3} = 7 - 3\ln 3$.

7. $v = 20 - 10t = 0$ at $t = 2$; rise $= \displaystyle\int_0^2 (20 - 10t)\,dt = 20$ metres.

8. Distance $\displaystyle\int_0^4 \left|e^{t/2}\cos t\right| dt \approx 9.515$; displacement ≈ -6.806.

9. (a) $v(t) = t(t-2)(t-4)$, which is negative for $2 < t < 4$. The particle moves left on that interval because $v < 0$.

   (b) With $s(t) = \dfrac{t^4}{4} - 2t^3 + 4t^2$: displacement on $[0,2]$ is 4 and on $[2,4]$ is -4. Position $x(4) = 3$; distance $4 + 4 = 8$.

   (c) $v(3) = -3$ and $a(3) = 3(9) - 12(3) + 8 = -1$. Both are negative, so the speed is increasing.

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